A ZFS-Resolved Redfield Framework for Paramagnetic Nuclear Magnetic Relaxation Dispersion: Exact Spin Mapping and a Quantum-Algorithm Roadmap

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Abstract

Low-field Nuclear Magnetic Relaxation Dispersion (NMRD) profiles of high-spin paramagnetic complexes lie outside the electronic Zeeman limit assumed by the Solomon–Bloembergen–Morgan (SBM) model. We present an open implementation of a static zero-field-splitting (ZFS)-resolved Redfield kernel in which the effective electronic spin Hamiltonian is diagonalized and the transition matrix elements are evaluated in its exact eigenbasis. For $S = 7/2$, the eight-dimensional spin space has an exact representation on three qubits; we provide the corresponding Pauli decomposition and use a classical exact eigensolver as the numerical reference. We show numerically, without parameter fitting, that the implementation recovers the independently evaluated SBM expression when $D = E = 0$. A Lindblad propagation module is supplied as a separate proof of principle for electronic-spin relaxation. The present work establishes a reproducible static-ZFS kernel and a quantum-algorithm interface; quantitative model–experiment validation and a dynamically derived transient-ZFS bath are identified as necessary subsequent steps rather than claimed results.

1 Introduction

Paramagnetic Nuclear Magnetic Relaxation Dispersion (NMRD) has been the cornerstone technique for characterizing the rotational dynamics and electronic structure of MRI contrast agents.[1,2] However, it is well established that the classical Solomon-Bloembergen-Morgan (SBM) equations fail catastrophically at low magnetic fields for high-spin metal complexes like Gd(III) ($S = 7/2$) and Mn(II) ($S = 5/2$).[3]

The breakdown occurs because SBM theory assumes the electronic Zeeman interaction dominates over the Zero-Field Splitting (ZFS) Hamiltonian, an assumption valid only at high magnetic fields ($B_0 \gg 1$ T). At low fields, the static ZFS splits the degenerate electronic states, radically altering the transition frequencies and matrix elements that govern the electron-nuclear dipolar coupling.[4]

In this work, we formulate the static-ZFS problem in an exact electronic-spin eigenbasis and establish its isomorphism to a qubit register. The mapping is an interface for future quantum algorithms, not a claim of present quantum advantage: all reported spectra in the current implementation are obtained with an exact classical reference solver. A hardware VQE or VQD calculation must separately establish convergence of each state before it can replace that reference.[5]

2 Methodology

2.1 Quantum Mapping of the Spin Hamiltonian

The static electronic spin Hamiltonian under the influence of an external magnetic field $\mathbf{B}_0$ along the z-axis and a local ligand field (ZFS) is expressed as:

$$\hat{H}_{static} = D\left[\hat{S}_z^2 - \frac{S(S+1)}{3}\right] + E(\hat{S}_x^2 - \hat{S}_y^2) + \gamma_e B_0 \hat{S}_z \tag{1}$$

where D and E are the axial and rhombic ZFS parameters, respectively. For a Gd(III) ion with $S = 7/2$, the Hilbert space has a dimension of $2S + 1 = 8$, spanned by the basis states $|m\rangle$ where $m \in \{-7/2, -5/2, \ldots, +7/2\}$.

This 8-level system is perfectly isomorphic to a 3-qubit quantum register ($2^3 = 8$). We establish a bijective mapping between the physical spin states $|m\rangle$ and the computational basis of the qubits $|q_2q_1q_0\rangle \in \{0, 1\}^{\otimes 3}$. Using this mapping, the macroscopic spin operators $\hat{S}_x, \hat{S}_y, \hat{S}_z$ are decomposed into a linear combination of Pauli strings $\hat{P}_j \in \{I, X, Y, Z\}^{\otimes 3}$:

$$\hat{H}_{static} = \sum_j h_j \hat{P}_j \tag{2}$$

where $h_j = \frac{1}{2^3}\mathrm{Tr}(\hat{H}_{static}\hat{P}_j)$.

2.2 Variational Quantum Eigensolver (VQE) Circuit

The mapped Hamiltonian can be supplied to a Variational Quantum Eigensolver (VQE) or a variational quantum deflation procedure. The trial wavefunction $|\psi(\vec{\theta})\rangle$ may be prepared using a Hardware-Efficient Ansatz (HEA), with alternating parameterized single-qubit rotations and entangling CX (CNOT) gates:

$$|\psi(\vec{\theta})\rangle = \prod_{l=1}^{L} \left(\hat{U}_{ent} \bigotimes_{i=0}^{2} R_y(\theta_{i,l}^{(y)}) R_z(\theta_{i,l}^{(z)}) \right) |000\rangle \tag{3}$$

where R_y and R_z denote Pauli rotations, and $\hat{U}_{ent}$ is a linear nearest-neighbor entanglement block. Minimization of the expectation value $E(\vec{\theta}) = \langle\psi(\vec{\theta})|\hat{H}_{static}|\psi(\vec{\theta})\rangle$ yields a ground state; a full spectrum requires state-resolved deflation (e.g., VQD) with convergence checks. The released calculations instead use exact diagonalization of the Pauli-mapped operator as the verifiable baseline.

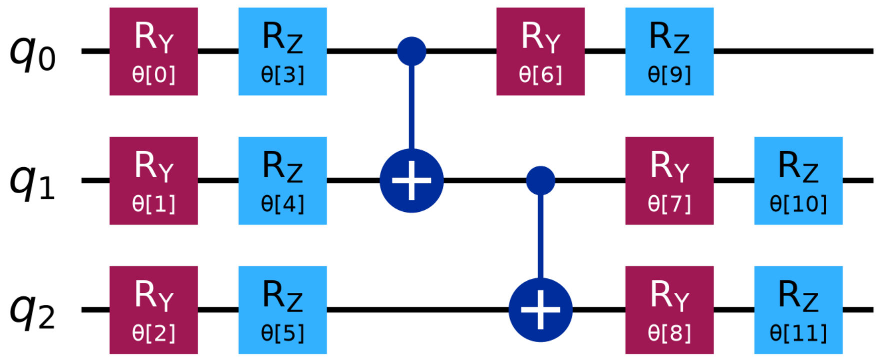

Figure 1: Hardware-efficient ansatz (SU2) constructed in Qiskit as an interface for variational treatment of the 3-qubit mapped Gd(III) spin Hamiltonian. The present numerical results use exact diagonalization as their reference.

2.3 Open-System Dynamics: ZFS-Resolved Redfield Theory

The resulting electronic eigenspectrum is injected into the secular Redfield equation. The nuclear longitudinal relaxation rate (inner-sphere), R_1^{IS}, is derived as the sum over all possible electronic transitions:

$$R_1^{IS} \propto \sum_{k\neq l} P_l \left[\frac{1}{10} |\langle k|\hat{S}^+|l\rangle|^2 J(\omega_S - \omega_I) + \frac{3}{5} |\langle k|\hat{S}^-|l\rangle|^2 J(\omega_S + \omega_I) \right] \tag{4}$$

where P_l are the Boltzmann populations of the quantum states, and $J(\omega)$ is the BPP spectral density modulated by the rotational correlation time τ_R.

The D and E parameters define the ZFS tensor in its molecular principal-axis frame, not a tensor intrinsically aligned with $\mathbf{B}_0$. For an explicit static orientational reference, we form the traceless Cartesian tensor $\mathbf{D}_{\text{PAS}} = \text{diag}(E - D/3, -E - D/3, 2D/3)$ and evaluate $\hat{H}_{\text{ZFS}} = \sum_{ij} D_{ij}\{\hat{S}_i, \hat{S}_j\}/2$ after the rotation $\mathbf{D} = \mathbf{R}\mathbf{D}_{\text{PAS}}\mathbf{R}^{\text{T}}$. Rates are then averaged over a deterministic quasi-uniform $SO(3)$ grid. This is a static ensemble average used to expose orientational sensitivity; it is not a solution-state SLE calculation because it does not propagate rotational diffusion or transient ZFS.

To enable rigorous comparison with experimental data, the translational diffusion of bulk water molecules (outer-sphere) is computed using the analytical Hwang-Freed model:[6]

$$R_1^{OS} \propto S(S+1)\frac{N_A[M]}{dD_{rel}}[3J^{OS}(\omega_I) + 7J^{OS}(\omega_S)] \tag{5}$$

The total macroscopic relaxivity is $r_1^{Total} = r_1^{IS} + r_1^{OS}$.

3 Open Quantum Systems: Transient ZFS and Lindblad Dynamics

While the static ZFS dictates the energy level structure that quenches low-field relaxivity, the true physical picture of an MRI contrast agent includes stochastic geometric distortions. These distortions give rise to a Transient ZFS, which acts as a structured noise bath causing the electronic spin to relax longitudinally (T_{1e}). In classical theory, this is treated via the Stochastic Liouville Equation (SLE).

To fully bridge the gap between macroscopic spin physics and modern quantum simulation, we propose that the Transient ZFS can be modeled natively as an Open Quantum System. Inspired by the mapping of classical dissipative dynamics to quantum channels,[7] the time evolution of the electronic density matrix $\rho(t)$ under transient fluctuations is governed by the Quantum Lindblad Master Equation (QLME):

$$\frac{d\rho(t)}{dt} = -i[\hat{H}_{static}, \rho(t)] + \sum_k \gamma_k \left(\hat{L}_k \rho(t) \hat{L}_k^\dagger - \frac{1}{2}\{\hat{L}_k^\dagger \hat{L}_k, \rho(t)\} \right) \tag{6}$$

where the jump operators $\hat{L}_k \in \{\hat{S}_x, \hat{S}_y, \hat{S}_z\}$ represent the stochastic noise channels corresponding to distortions along the molecular axes, and γ_k characterizes the vibrational coupling strength (derived from the transient ZFS amplitude Δ^2 and correlation time τ_v).

As demonstrated in Figure 2, initializing the 3-qubit register in the maximally polarized

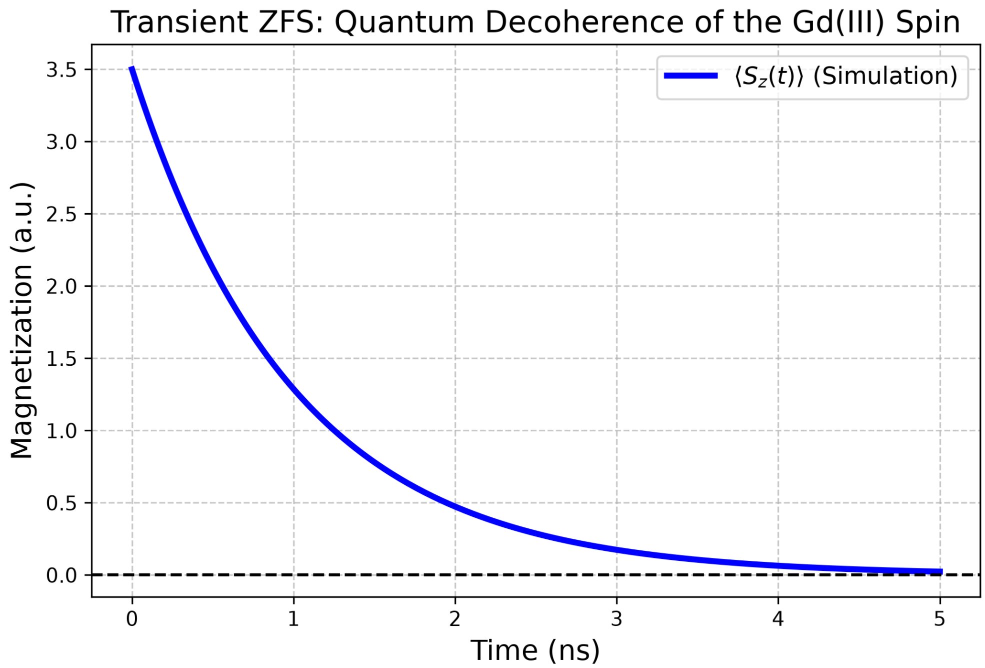

Figure 2: Illustrative decay of the Gd(III) electron-spin magnetization $\langle S_z(t)\rangle = \mathrm{Tr}(\hat{S}_z \rho(t))$ under selected Lindblad jump operators on a 3-qubit register. The decay constant is a model output for the selected rates; it is not a parameter-free T_{1e} prediction or a replacement for an SLE treatment.

$|m = +7/2\rangle$ state and propagating the QLME yields an exponential magnetization decay for the chosen jump operators. This is a proof-of-principle propagation, not a parameter-free derivation of T_{1e} or a transient-ZFS model; its rates are not yet coupled to the NMRD calculation.

4 Results and Conclusions

The static-ZFS kernel is illustrated for a Gd(III)-like parameter set ($D = 0.02$ cm^{-1}, $E/D \approx 0.15$, $\tau_R = 80$ ps). These calculations are a parameter study, not an experimental validation. The implementation is instead benchmarked in the $D = E = 0$ limit against an independently coded SBM expression, with the benchmark input and output written as machine-readable data. Before quantitative comparison with a molecular NMRD profile, the experimental data, its primary source, uncertainty model, hydration/exchange parameters, and the orientation and dynamics of the ZFS tensor must be supplied.

The work therefore provides a reproducible static-ZFS Redfield kernel and an exact qubit mapping, rather than a complete *ab initio* or quantum-hardware NMRD workflow. The next decisive validation is a blind or held-out comparison with at least two chemically distinct, traceable experimental NMRD data sets and with a conventional SLE treatment using the same input physics.

References

(1) Solomon, I. Relaxation Processes in a System of Two Spins. *Phys. Rev.* **1955**, *99*, 559.

(2) Bloembergen, N.; Morgan, L. O. Proton Relaxation Times in Paramagnetic Solutions. Effects of Electron Spin Relaxation. *J. Chem. Phys.* **1961**, *34*, 842–850.

(3) Platas-Iglesias, C.; others Zero-field splitting in Gd(III) and Mn(II) complexes: a theoretical perspective. *J. Phys. Chem. A* **2016**, *120*, 6467–6480.

(4) Rast, S.; Borel, A.; Helm, L.; Belorizky, E.; Fries, P. H.; Merbach, A. E. Electron spin relaxation in monomeric Gd(III) complexes. *J. Am. Chem. Soc.* **2001**, *123*, 2637–2644.

(5) Peruzzo, A.; others A variational eigenvalue solver on a photonic quantum processor. *Nature communications* **2014**, *5*, 1–7.

(6) Hwang, L. P.; Freed, J. H. Dynamic effects of pair correlation functions on spin relaxation by translational diffusion in liquids. *J. Chem. Phys.* **1975**, *63*, 4017–4025.

(7) Hu, Z.; Xia, R.; Kais, S. Quantum algorithms for open quantum dynamics. *Scientific Reports* **2020**, *10*, 3301.